DBMS Mini-Project

AutoCare: Vehicle Service & Maintenance Management System

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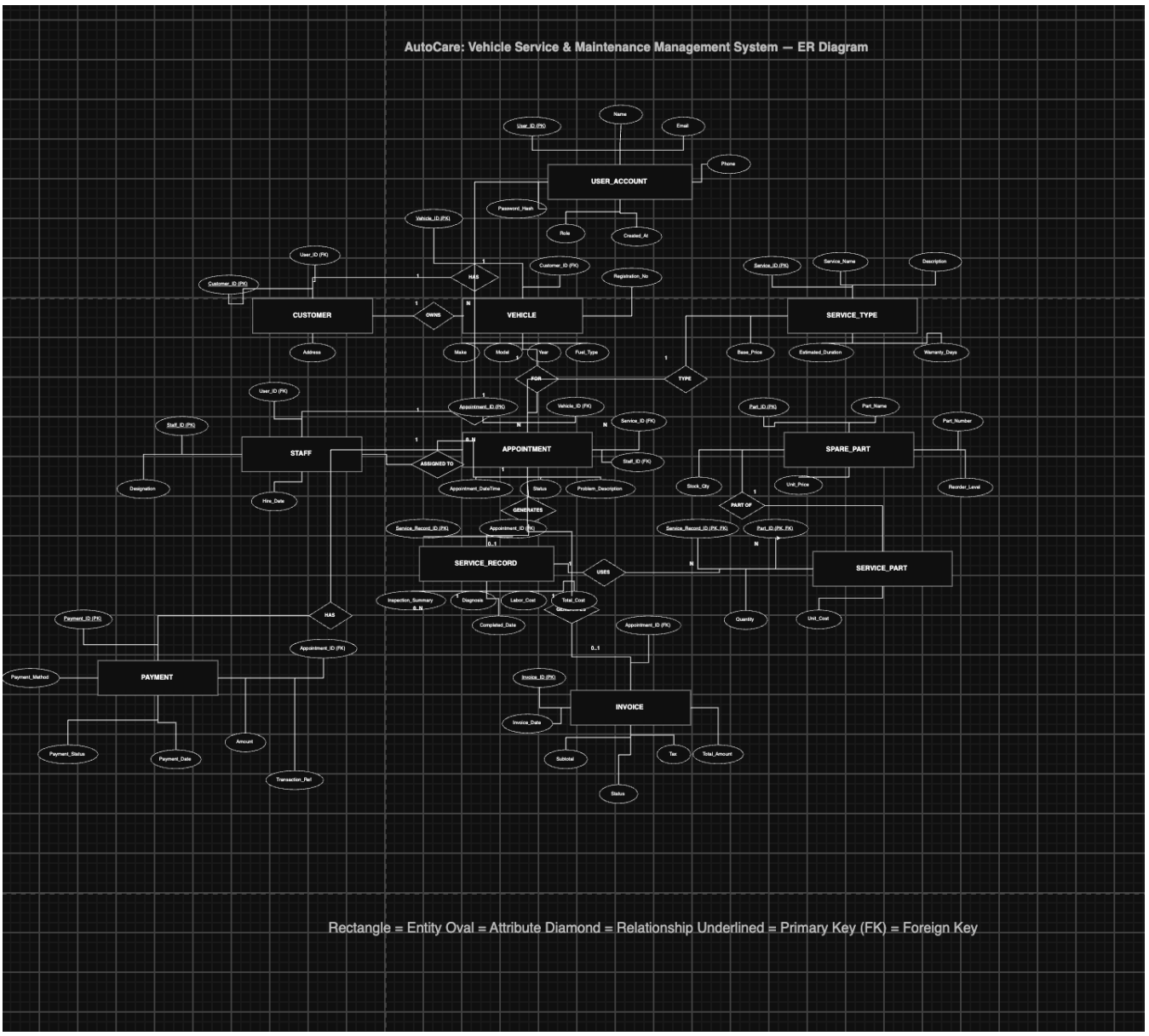
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Project Title

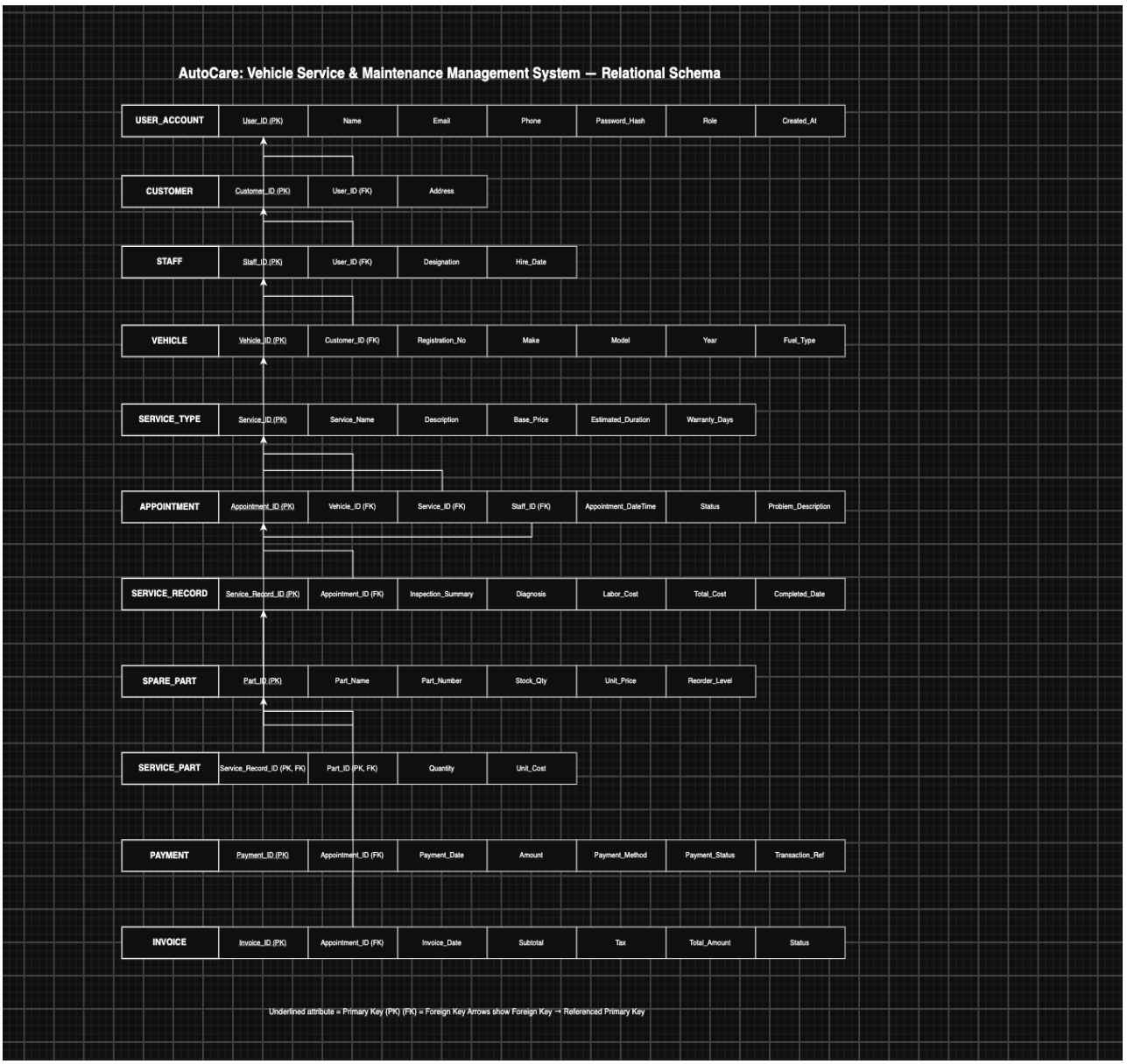
AutoCare: Vehicle Service & Maintenance Management System

AutoCare is a database-driven vehicle service and maintenance management system that manages customers, vehicles, staff, service appointments, service records, spare parts, payments, and invoices.

ER Diagram



Schema Diagram


AutoCare: Vehicle Service & Maintenance Management System — Relational Schema
USER_ACCOUNT
User_ID (PK)
Name
Email
Phone
Password_Hash
Role
Created_At
CUSTOMER
Customer_ID (PK)
User_ID (FK)
Address
STAFF
Staff_ID (PK)
User_ID (FK)
Designation
Hire_Date
VEHICLE
Vehicle_ID (PK)
Customer_ID (FK)
Registration_No
Make
Model
Year
Fuel_Type
SERVICE_TYPE
Service_ID (PK)
Service_Name
Description
Base_Price
Estimated_Duration
Warranty_Days
APPOINTMENT
Appointment_ID (PK)
Vehicle_ID (FK)
Service_ID (FK)
Staff_ID (FK)
Appointment_DateTime
Status
Problem_Description
SERVICE_RECORD
Service_Record_ID (PK)
Appointment_ID (FK)
Inspection_Summary
Diagnosis
Labor_Cost
Total_Cost
Completed_Date
SPARE_PART
Part_ID (PK)
Part_Name
Part_Number
Stock_Qty
Unit_Price
Reorder_Level
SERVICE_PART
Service_Record_ID (PK, FK)
Part_ID (PK, FK)
Quantity
Unit_Cost
PAYMENT
Payment_ID (PK)
Appointment_ID (FK)
Payment_Date
Amount
Payment_Method
Payment_Status
Transaction_Ref
INVOICE
Invoice_ID (PK)
Appointment_ID (FK)
Invoice_Date
Subtotal
Tax
Total_Amount
Status
Underlined attribute = Primary Key (PK) (FK) = Foreign Key Arrows show Foreign Key → Referenced Primary Key